

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

Name:P Madhavi

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COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

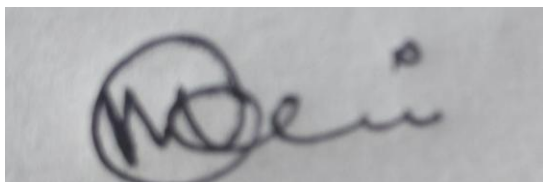
Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I P Madhavi(192372133) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in dark ink on a light-colored surface. The signature is stylized, starting with a large, circular initial that appears to be 'P' or 'M', followed by a series of loops and a final flourish that ends in a small dot.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

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① Responsive web design concept and Importance:

* Approach to web design ensuring content renders well on various screen sizes

* Importance: Enhances, usability, improves mobile SEO rankings, and reduces bounce rates across device.

Key CSS Techniques:-

* Media Queries: Apply distinct CSS rule based on device width or resolution.

flexible units: use relative units like percentages, vw, or vh instead of fixed pixels

flexible images: set max-width: 100% to prevent images from overflowing containers.

Media Queries: A CSS3 feature that applies specific style blocks only if the device meets certain conditions (e.g., @media (max-width: 768px) block styles specifically for mobile devices).

Adaptive layouts: flexbox vs. Grid

CSS flexbox: A one-dimensional layout model. It aligns items in either a single row or single column. It automatically expands or shrinks item to fill available space, making it perfect for navigation bars or simple card lists.

CSS Grid: A two-dimensional layout model. It manages both rows and columns simultaneously. It is ideal for complex, master-pages layouts where elements need precise alignments in both directions.

Simple Example layout:

```
• container {  
  display: flex;  
  flex-wrap: wrap  
}
```

```
• card {  
  flex: 1 1 30%;  
  margin: 10px;  
}
```

```
@media (max-width: 600px) {
```

```
• card {  
  flex: 1 1 100%;  
}
```

Suggested user experience

* Touch-target organization

* Typography Scaling.

② Concept and purpose of client-side validation:-

concept: Client-side validation is the process of checking user input for errors or missing data directly in the web browser before the form is sent to the server.

Purpose: It provides immediate feedback to the user, catches formatting mistakes instantly, and reduces necessary server load by looking invalid data requests from traveling over the network.

2. JavaScript Event handling in validation

* `Onsubmit` or `addEventListener('submit')`: used on the form element to intercept the submission process; if validation fails, `event.preventDefault()` is called to stop the form sending data.

Onblur or addEventListener Triggers validation immediately when a user clicks or moves away from a specific input field, allowing real-time error corrections.

Role of Regular Expressions:

Definition: A regular expression is a sequence of characters that forms a search pattern used to match search, and validate specific text formats.

Email Validation: used patterns to ensure the input contains a local part, an @ symbol a domain name and a valid extension.

Phone number validation Enforces numerical constraints, specific lengths and optional characters like country codes, dashes, or parentheses.

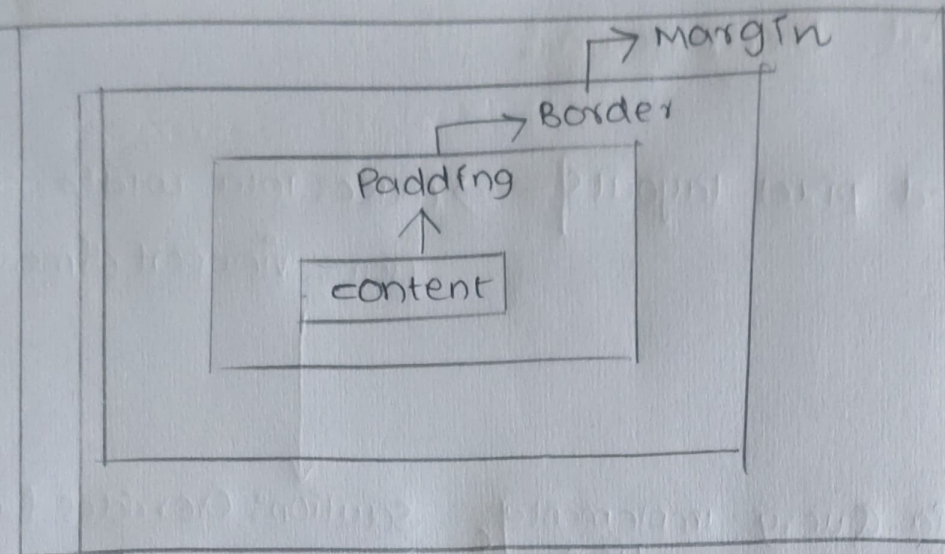
4. Suitable examples

```
<form id = "myform">  
<input type = "text" id = "emailfield" />  
<span id = "errorMsg" style = "color: red;" > </span>  
<button type = "submit"> Submit </button>  
</form>  
<script>  
document.getElementById('myform').addEventListener  
'submit', function (event) {
```

```
    constant email = document.getElementById('emailfield')  
} - - - -  
5. suggested usability and security improvements
```

- * usability (inline error messaging)
- * usability (HTML5 attributes)
- * Security (Mandatory server-side validation).

3



1. The CSS Box model and components

Every visible HTML element is a box composed of four concentric layers

* content: The core area where text, images, or child elements reside.

* padding: The clear transparent space surrounding the content, located inside the border.

* border: The structural edge or line wrapping around the padding and content.

* margin: The outermost layer transparent space separating the element from its neighbouring elements.

How CSS positioning affects layout structure?

* Static

└ Absolute

└ fixed/sticky

3. layout Break down:

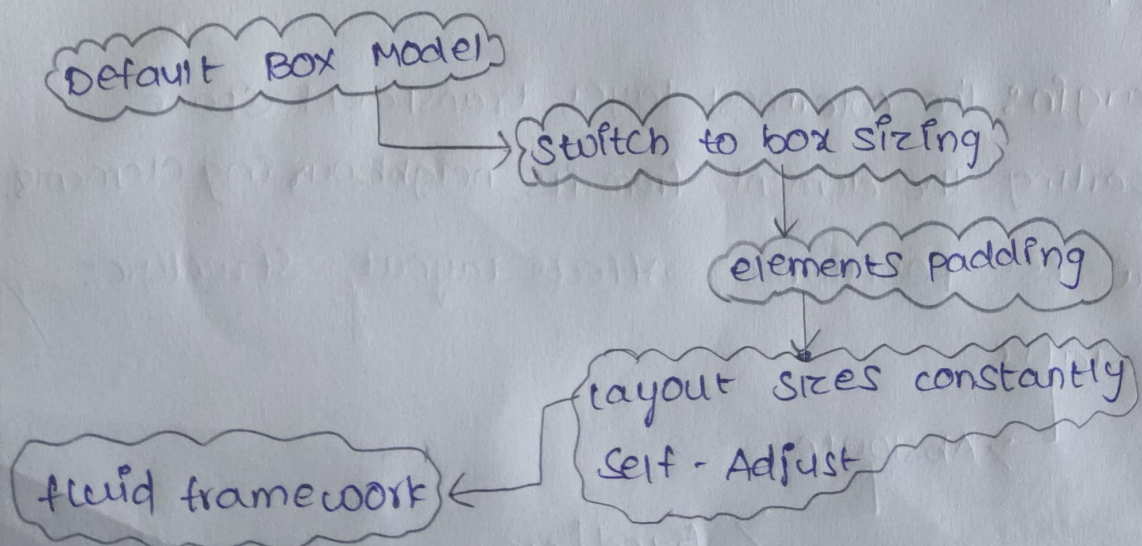
[fixed pixel layouts] → cause: Total width overflows viewport dimensions

[Media Query implementation] ← Solution: Overrides fixed values at custom break points

Possible causes of layout Breaking

- * using fixed pixel width (like width 1000px)
- * forgetting the default content-box calculation.
- * using media queries to resolve.
- * The apply tailored CSS rules using conditional checks like @media (max-width: 768px)

layout Stabilization



4) concept of the document object model (DOM) and its role?

* The DOM is a programming interface for web documents. It represents the structure of HTML pages as a hierarchical trees of objects (models), where every element that programming languages can access.

Role: It acts a bridge between the static web page and programming languages like JavaScript.

How JavaScript manipulates the DOM dynamically

* selecting elements

* Modifying content & styles

* Handling Events & Networks Requests

Suitable examples:

```
<!-- HTML -->
```

```
<div id = "new container">
```

```
<p id = "new-text">
```

```
</div>
```

```
<script>
```

```
const loadButton = document.getElementById('load-btn')
```

```
const newText = document.getElementById('new-text')
```



```
load Button.addEventListener('click')  
newsText.textContent = "loading updates"
```

```
setTimeout(() => {
```

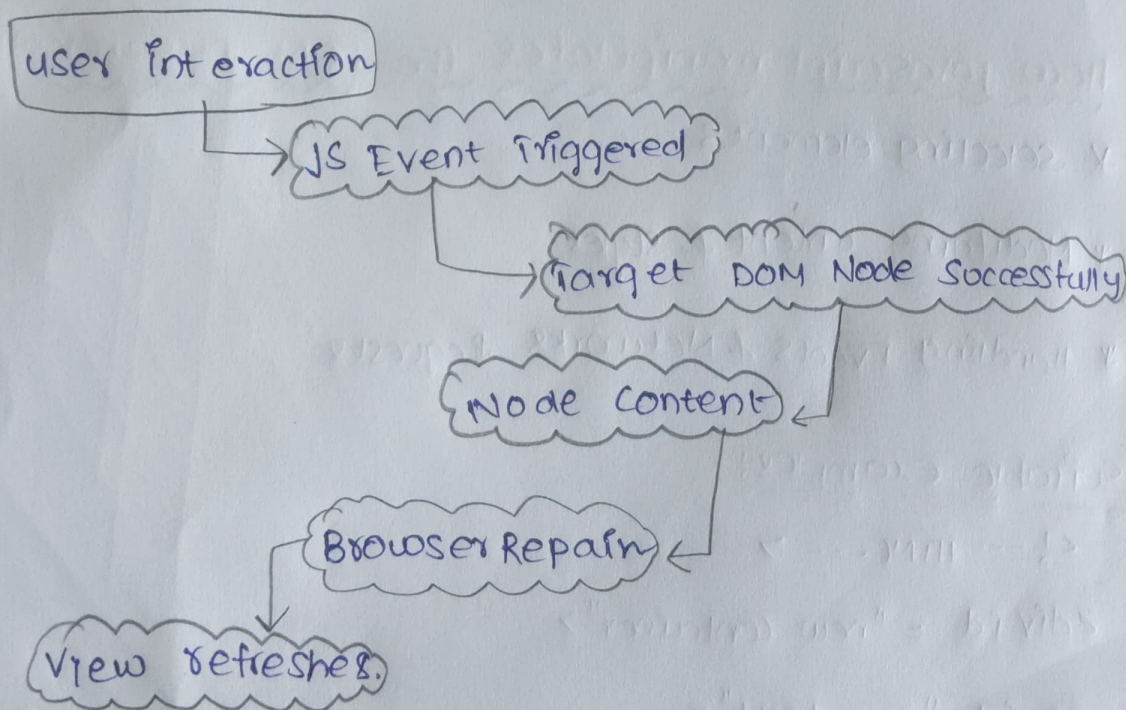
```
  newsText.textContent = "Breaking News: The DOM been  
  updated successfully without a page"
```

```
}, 1000);
```

```
} }
```

```
</script>
```

DOM updates flowchart



Suggested Performances and ux Improvements

* performance (Batch DOM changes)

* performance (Debouncing & Throttling)